

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Project Title

E-Commerce Order Management Database System

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Project Title:

E-Commerce Order Management Database System

Course: Database Management System

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1. Introduction

1.1 Purpose

The purpose of this Software Requirement Specification (SRS) is to define the functional and non-functional requirements of the E-Commerce Order Management Database System. The document serves as a blueprint for designing, developing, testing, and maintaining the database system.

1.2 Scope

The system provides centralized management of customers, products, suppliers, orders, payments, shipments, and customer reviews. It improves operational efficiency, reduces data redundancy, and supports business reporting.

1.3 Objectives

- Centralize business data
 - Improve customer service
 - Automate order processing
 - Manage inventory efficiently
 - Secure payment records
 - Track deliveries
 - Generate business reports
-

1.4 Intended Users

- Administrator

- Customer
 - Supplier
 - Warehouse Staff
 - Finance Staff
 - Delivery Staff
-

1.5 Definitions

Term	Description
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Customer	Person purchasing products
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Supplier	Company supplying products
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Order	Customer purchase request
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Shipment	Delivery information
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Payment	Financial transaction
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Review	Customer feedback
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2. Overall Description

2.1 Product Perspective

The system is a centralized relational database supporting all major e-commerce operations.

2.2 Product Functions

The system shall:

- Register customers
- Manage products
- Manage suppliers
- Process orders
- Record payments

- Track shipments
 - Store reviews
 - Generate reports
-

2.3 User Classes

Administrator

- Manage entire database
- View reports
- Manage users

Customer

- Register
- Place orders
- Review products

Supplier

- Supply products
- Update product availability

Warehouse Staff

- Update stock

Delivery Staff

- Update shipment status

Finance Staff

- Verify payments
-

2.4 Operating Environment

- Windows 10/11
- Linux Server
- MySQL Database
- Java Application

- Internet Connection
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2.5 Assumptions

- Internet connectivity is available.
 - Users have valid login credentials.
 - Products exist before orders are placed.
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2.6 Constraints

- Only specified entities are used.
 - Data must remain consistent.
 - Database follows relational design principles.
-

3. Functional Requirements

FR-01 Customer Management

- Customer Registration
 - Update Profile
 - View Profile
-

FR-02 Category Management

- Add Category
 - Update Category
 - Delete Category
-

FR-03 Product Management

- Add Product
- Update Product
- Delete Product
- View Product

FR-04 Supplier Management

- Add Supplier
- Update Supplier
- Delete Supplier

FR-05 Order Management

- Create Order
- Cancel Order
- Update Order Status
- View Order History

FR-06 Order Details

- Add Products
- Update Quantity
- Calculate Total

FR-07 Payment Management

- Record Payment
- Verify Payment
- Update Payment Status

FR-08 Shipment Management

- Create Shipment
- Update Delivery Status
- Track Shipment

FR-09 Review Management

- Submit Review
 - View Reviews
 - Manage Ratings
-

4. Non-Functional Requirements

Performance

- Response time less than 3 seconds.
 - Support multiple simultaneous users.
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Security

- User Authentication
 - Role-Based Access Control
 - Password Encryption
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Reliability

- Data Accuracy
 - Transaction Integrity
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Availability

- System available 24×7.
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Scalability

- Support increasing users and products.
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Maintainability

- Easy updates and modifications.
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Backup and Recovery

- Automatic database backup.
 - Data recovery after failure.
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5. External Interface Requirements

User Interface

- Simple navigation
 - Responsive design
 - Search functionality
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Hardware Interface

- Desktop Computer
 - Laptop
 - Server
-

Software Interface

- MySQL
 - Java
 - Windows/Linux
-

Communication Interface

- Internet
 - TCP/IP
-

6. Database Requirements

Core Entities

Customer

- Customer ID
- Name

- Email
 - Mobile Number
 - Address
 - Registration Date
-

Category

- Category ID
 - Category Name
 - Description
-

Product

- Product ID
 - Product Name
 - Price
 - Stock Quantity
 - Category ID
-

Supplier

- Supplier ID
 - Supplier Name
 - Contact Information
-

Order

- Order ID
- Customer ID
- Order Date
- Total Amount
- Order Status

Order Details

- Order Detail ID
- Order ID
- Product ID
- Quantity
- Unit Price

Payment

- Payment ID
- Order ID
- Payment Method
- Payment Date
- Payment Status

Shipment

- Shipment ID
- Order ID
- Delivery Address
- Shipment Date
- Delivery Status

Review

- Review ID
- Customer ID
- Product ID
- Rating
- Comments

7. Business Rules

- Customer ID must be unique.
- Product belongs to one category.
- Stock quantity cannot be negative.
- Orders require at least one product.
- Payment must be completed before shipment.
- Ratings range from 1 to 5.
- Reviews can only be submitted by registered customers.

8. Security Requirements

- User Login Authentication
- Password Encryption
- Data Backup
- Database Recovery
- Audit Logs
- Secure Payment Information
- Role-Based Access Control

9. Assumptions and Constraints

Assumptions

- Internet access is available.
- Users have valid credentials.
- Products are available in inventory.

Constraints

- Only specified entities are used.
- No entity attributes may be modified.
- Relational database design only.

10. Future Enhancements

- Mobile Application
- AI Product Recommendation
- Online Payment Gateway Integration
- Sales Analytics Dashboard
- Multi-language Support
- Barcode Integration

11. Conclusion

The E-Commerce Order Management Database System provides a centralized and efficient solution for managing customer information, products, suppliers, orders, payments, shipments, and customer reviews. The system is designed to reduce manual effort, improve operational efficiency, maintain data consistency, and support informed decision-making through accurate reporting. This Software Requirement Specification serves as the foundation for the design, development, implementation, and testing of the database system.

Appendix